

# 911 Emergency Dispatch System Design Document

## City Map & Officer Sector Mapping

The system monitors 10 distinct municipal landmark locations across New York City. Each landmark is assigned a unique location code (L01 through L10) and mapped directly to a primary response sector from the 60-officer database.

### Location Code & Sector Mapping

| Location Code | City Landmark / Location                | Primary Response Sector | Base Officer Allocation       |
|---------------|-----------------------------------------|-------------------------|-------------------------------|
| L01           | Times Square                            | Manhattan               | 2 Police, 2 Paramedic, 2 Fire |
| L02           | Washington Square Park                  | Lower East Side         | 2 Police, 2 Paramedic, 2 Fire |
| L03           | Union Square                            | Lower East Side         | Shared Lower East Side Units  |
| L04           | Brooklyn Bridge                         | Brooklyn                | 2 Police, 2 Paramedic, 2 Fire |
| L05           | Coney Island                            | Brooklyn                | Shared Brooklyn Units         |
| L06           | Central Park<br>(Mall/Bethesda Terrace) | Upper West Side         | 2 Police, 2 Paramedic, 2 Fire |
| L07           | Williamsburg, Brooklyn                  | Brooklyn                | Shared Brooklyn Units         |
| L08           | SoHo                                    | Lower East Side         | Shared Lower East Side Units  |
| L09           | Bryant Park                             | Manhattan               | Shared Manhattan Units        |
| L10           | East Village                            | Lower East Side         | Shared Lower East Side Units  |

**Rank Fallback Rule:** If no officer of the required rank is currently available in the call's primary sector, the system performs a city-wide search by rank, breaking ties by lowest active case count, then officer ID. This guarantees every severity level is dispatchable from every location.

### Complete Officer Database Registry (60 Officers)

The system maintains a local registry of 60 officers across 10 sectors (6 officers per sector: 2 Police, 2 Paramedic, 2 Fire Department).

| ID    | Name          | Department | Sector Location | Rank             |
|-------|---------------|------------|-----------------|------------------|
| P-101 | Maria Chen    | Police     | Manhattan       | Patrol Officer   |
| P-102 | David Okafor  | Police     | Manhattan       | Sergeant         |
| M-101 | Sarah Wilson  | Paramedic  | Manhattan       | EMT-Basic        |
| M-102 | Thomas Reed   | Paramedic  | Manhattan       | Paramedic        |
| F-101 | Tom Harris    | Fire       | Manhattan       | Firefighter      |
| F-102 | Ken Lee       | Fire       | Manhattan       | Lieutenant       |
| P-103 | Anna Martinez | Police     | Brooklyn        | Lieutenant       |
| P-104 | James Carter  | Police     | Brooklyn        | Commander        |
| M-103 | Neha Patel    | Paramedic  | Brooklyn        | EMT-Basic        |
| M-104 | Robert Smith  | Paramedic  | Brooklyn        | Senior Paramedic |

| ID    | Name          | Department | Sector Location | Rank             |
|-------|---------------|------------|-----------------|------------------|
| F-103 | Elena Vasquez | Fire       | Brooklyn        | Captain          |
| F-104 | Laura Brown   | Fire       | Brooklyn        | Battalion Chief  |
| P-105 | Daniel Kim    | Police     | Queens          | Patrol Officer   |
| P-106 | Olivia Davis  | Police     | Queens          | Sergeant         |
| M-105 | Michael Brown | Paramedic  | Queens          | Paramedic        |
| M-106 | Priya Shah    | Paramedic  | Queens          | Senior Paramedic |
| F-105 | Alex Johnson  | Fire       | Queens          | Firefighter      |
| F-106 | Ryan Cooper   | Fire       | Queens          | Lieutenant       |
| P-107 | Ethan Wilson  | Police     | Bronx           | Sergeant         |
| P-108 | Sophia Garcia | Police     | Bronx           | Lieutenant       |
| M-107 | James Miller  | Paramedic  | Bronx           | EMT-Basic        |
| M-108 | Aisha Khan    | Paramedic  | Bronx           | Paramedic        |
| F-107 | Daniel Clark  | Fire       | Bronx           | Lieutenant       |
| F-108 | Marcus Taylor | Fire       | Bronx           | Captain          |
| P-109 | Noah Anderson | Police     | Staten Island   | Patrol Officer   |

| ID    | Name           | Department | Sector Location | Rank             |
|-------|----------------|------------|-----------------|------------------|
| P-110 | Emma Thomas    | Police     | Staten Island   | Sergeant         |
| M-109 | Liam Moore     | Paramedic  | Staten Island   | EMT-Basic        |
| M-110 | Grace Martin   | Paramedic  | Staten Island   | Senior Paramedic |
| F-109 | Henry Jackson  | Fire       | Staten Island   | Firefighter      |
| F-110 | Chloe White    | Fire       | Staten Island   | Lieutenant       |
| P-111 | William Harris | Police     | Harlem          | Sergeant         |
| P-112 | Mia Thompson   | Police     | Harlem          | Lieutenant       |
| M-111 | Benjamin Young | Paramedic  | Harlem          | EMT-Basic        |
| M-112 | Ava Walker     | Paramedic  | Harlem          | Paramedic        |
| F-111 | Samuel Hall    | Fire       | Harlem          | Firefighter      |
| F-112 | Lily Allen     | Fire       | Harlem          | Captain          |
| P-113 | Ethan Wang     | Police     | Flushing        | Patrol Officer   |
| P-114 | Isabella King  | Police     | Flushing        | Sergeant         |
| M-113 | Daniel Wright  | Paramedic  | Flushing        | EMT-Basic        |
| M-114 | Sophia Lewis   | Paramedic  | Flushing        | Senior Paramedic |

| ID    | Name              | Department | Sector Location | Rank             |
|-------|-------------------|------------|-----------------|------------------|
| F-113 | Jacob Scott       | Fire       | Flushing        | Firefighter      |
| F-114 | Emily Green       | Fire       | Flushing        | Lieutenant       |
| P-115 | Matthew Baker     | Police     | Astoria         | Lieutenant       |
| P-116 | Amelia Adams      | Police     | Astoria         | Sergeant         |
| M-115 | Lucas Nelson      | Paramedic  | Astoria         | Paramedic        |
| M-116 | Hannah Hill       | Paramedic  | Astoria         | Senior Paramedic |
| F-115 | Mason Campbell    | Fire       | Astoria         | Firefighter      |
| F-116 | Evelyn Mitchell   | Fire       | Astoria         | Captain          |
| P-117 | Alexander Roberts | Police     | Lower East Side | Patrol Officer   |
| P-118 | Charlotte Turner  | Police     | Lower East Side | Lieutenant       |
| M-117 | Jack Phillips     | Paramedic  | Lower East Side | EMT-Basic        |
| M-118 | Sofia Evans       | Paramedic  | Lower East Side | Paramedic        |
| F-117 | Leo Parker        | Fire       | Lower East Side | Lieutenant       |
| F-118 | Ella Edwards      | Fire       | Lower East Side | Battalion Chief  |
| P-119 | Henry Collins     | Police     | Upper West Side | Sergeant         |

| ID    | Name           | Department | Sector Location | Rank        |
|-------|----------------|------------|-----------------|-------------|
| P-120 | Amelia Stewart | Police     | Upper West Side | Commander   |
| M-119 | Oliver Sanchez | Paramedic  | Upper West Side | EMT-Basic   |
| M-120 | Harper Morris  | Paramedic  | Upper West Side | EMT LEAD    |
| F-119 | William Rogers | Fire       | Upper West Side | Firefighter |
| F-120 | Abigail Reed   | Fire       | Upper West Side | Captain     |

Note: Base Officer Allocation refers to lead officers stationed per sector. Required personnel (Severity  $\times$  2) counts total responding personnel including support staff, who are tallied in the dispatch decision but not individually assigned from the registry.

## Terminology & Formulae

### Key Terminology

- **Call ID:** Unique alphanumeric identifier generated sequentially (CALL-001, CALL-002, etc.).
- **Severity Rating (S):** Integer value  $S \in \{1,2,3,4,5\}$  representing incident urgency.
- **Unit Status:** State lifecycle of a responding unit: ASSIGNED  $\rightarrow$  EN\_ROUTE  $\rightarrow$  ON\_SCENE  $\rightarrow$  CLEARED.
- **Call Status:** State lifecycle of an emergency call: PENDING  $\rightarrow$  COMPLETED.
- **Waitlist:** Holding list for calls logged when every officer of a department is at capacity ( $\geq 3$  active cases). Ordered by severity (descending), then arrival. Re-dispatched automatically after each completion.

### Mathematical Formulae & Allocation Logic

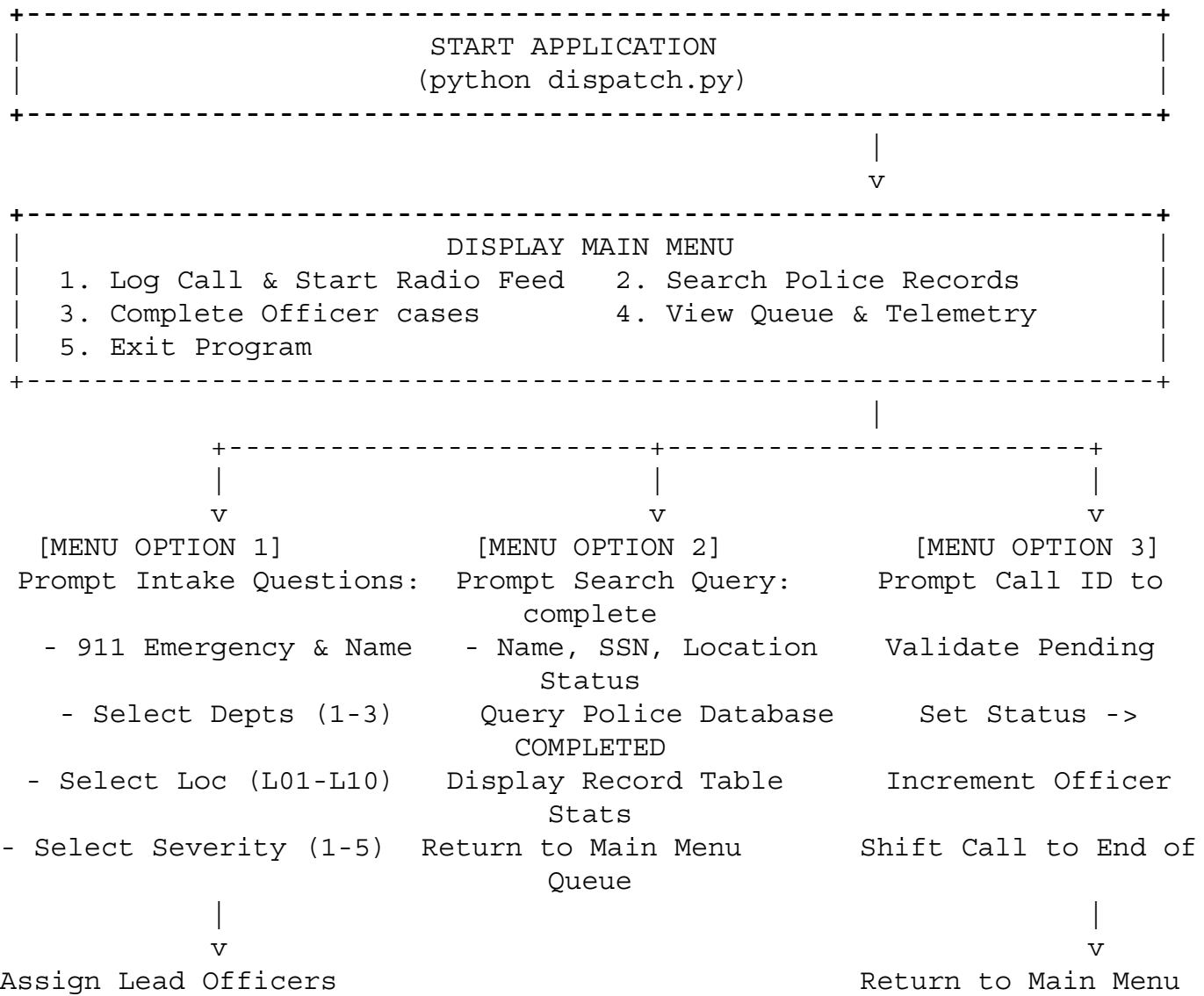
1. **Required Personnel Calculation:** The number of assigned support personnel per selected department ( $P_{dept}$ ) is directly proportional to severity (S):  

$$P_{dept} = S \times 2 \text{ where } S \in [1,5]$$
2. **Lead Rank Selection Matrix:** Lead officer rank is dynamically determined based on Severity Level (S):

| Severity Band   | Police         | Paramedic        | Fire            |
|-----------------|----------------|------------------|-----------------|
| S = 1 -2(Low)   | Patrol officer | EMT - Basic      | Firefighter     |
| S = 3(Medium)   | Sergeant       | Paramedic        | Lieutenant      |
| S = 4(High)     | Lieutenant     | Senior Paramedic | Caption         |
| S = 5(Critical) | Commander      | EMT - Lead       | Battalion Chief |

3. **Queue Sorting Comparator:** Calls in the global queue (Q) are sorted using a tuple priority key  $K(\text{call})$  to guarantee pending calls stay prioritized at the top:
4.  $K(\text{call}) = (\text{IsCompleted}(\text{call}), -\text{Severity}(\text{call}), \text{SequenceID}(\text{call}))$   
where  $\text{IsCompleted} = 0$  for Pending / 1 for Completed, and  $\text{SequenceID}$  is the integer creation order — not the CALL-XXX string.

## Program Flow



```

Start Interactive Radio Feed:
-> Unit Accept ("EN_ROUTE")
-> Unit Arrive ("ON_SCENE")
-> Prompt Completion (Y/N)
    |
+-----+-----+
    |               |
    v               v
[MENU OPTION 4]    [MENU OPTION 5]
Render Dual Tables: Any severity -5 calls
    pending
    1. Global Call Queue - Y:[warning]Critical
        severity
    2. Officer Telemetry 5 calls are still
        pending
Return to Main Menu    confirm exit?(Y/N)
                        N -> Return to main
                        menu
                        - N:continue
                        Export
                        Shift_summary.txt|save
                        telemetry| Exit

```

## Data Structures

```

from dataclasses import dataclass, field
from enum import Enum
from typing import List, Dict, Optional

```

```

class ServiceType(Enum):
    POLICE = "Police"
    FIRE = "Fire"
    PARAMEDIC = "Paramedic"

```

```

class CallStatus(Enum):
    PENDING = "Pending"
    COMPLETED = "Completed"

```

```

class UnitStatus(Enum):
    ASSIGNED = "Assigned"
    EN_ROUTE = "En Route"
    ON_SCENE = "On Scene"
    CLEARED = "Cleared"

```

```

@dataclass
class Officer:
    officer_id: str

```

```

name: str
department: ServiceType
location_sector: str
rank: str
active_cases: int = 0
completed_cases: int = 0

@dataclass
class PoliceRecord:
    record_id: str
    person_name: str
    ssn_last4: str
    warrants: List[str]
    citations: List[str]
    prior_incidents: List[str]

@dataclass
class EmergencyCall:
    call_id: str
    caller_name: str
    incident_description: str
    departments: List[ServiceType]
    location_code: str
    landmark_name: str
    severity: int
    personnel_per_dept: int
    assigned_officers: Dict[ServiceType, Officer]
    unit_status: UnitStatus = UnitStatus.ASSIGNED
    call_status: CallStatus = CallStatus.PENDING

class DispatchSystemState:
    def __init__(self):
        self.call_queue: List[EmergencyCall] = []
        self.officer_registry: Dict[str, Officer] = {}
        self.police_database: List[PoliceRecord] = []
        self.call_counter: int = 1

```

## Function Signatures

| Function Signature                                   | Description                                                  |
|------------------------------------------------------|--------------------------------------------------------------|
| def initialize_system_data() -> DispatchSystemState: | Instantiates state and loads 60 officers and police records. |

| Function Signature                                                                                                                                                   | Description                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <code>def calculate_personnel_count(severity: int) -&gt; int:</code>                                                                                                 | Returns required personnel count (Severity×2).                                   |
| <code>def select_lead_officers(state: DispatchSystemState, depts: List[ServiceType], location: str, severity: int) -&gt; Dict[ServiceType, Officer]:</code>          | Selects available lead officer per department matching rank and location sector. |
| <code>def log_emergency_call(state: DispatchSystemState, caller: str, desc: str, depts: List[ServiceType], loc_code: str, severity: int) -&gt; EmergencyCall:</code> | Validates input, constructs EmergencyCall, and appends to active queue.          |
| <code>def run_radio_dialogue_feed(call: EmergencyCall) -&gt; bool:</code>                                                                                            | Executes real-time terminal radio conversation loop between operator and unit.   |
| <code>def complete_call(state: DispatchSystemState, call_id: str) -&gt; bool:</code>                                                                                 | Marks call completed, updates officer telemetry, and re-orders queue to bottom.  |
| <code>def search_police_records(state: DispatchSystemState, query: str) -&gt; List[PoliceRecord]:</code>                                                             | Filters local police database records matching query string.                     |
| <code>def sort_queue(state: DispatchSystemState) -&gt; None:</code>                                                                                                  | Sorts queue placing Pending calls at top and Completed calls at bottom.          |
| <code>def display_queue_and_telemetry(state: DispatchSystemState) -&gt; None:</code>                                                                                 | Formats and prints Markdown/ASCII tables for queue and officer stats.            |

## Function-Level Algorithms

### **log\_emergency\_call(state, caller, desc, depts, loc\_code, severity)**

1. Calculate required personnel:  $P = \text{calculate\_personnel\_count}(\text{severity})$ .
2. Generate sequential call\_id: "CALL-" + str(state.call\_counter).zfill(3).
3. Resolve landmark\_name from loc\_code mapping table.
4. Execute select\_lead\_officers(state, depts, landmark\_name, severity) to get assigned officers.
5. Construct new EmergencyCall instance with call\_status = CallStatus.PENDING.

6. Increment `active_cases` by 1 for each assigned lead officer.
7. Append call object to `state.call_queue`.
8. Increment `state.call_counter` by 1.
9. Return the created `EmergencyCall`.

### **run\_radio\_dialogue\_feed(call)**

1. Get the lead officer assigned to the first department in `call.departments`.
2. Output system log: "[RADIO FEED] " + `officer.rank` + " " + `officer.name` + ": 'We are accepting the case and heading to location.'"
3. Update `call.unit_status = UnitStatus.EN_ROUTE`.
4. Output system log: "[RADIO FEED] " + `officer.rank` + " " + `officer.name` + ": 'We have reached the location.'"
5. Update `call.unit_status = UnitStatus.ON_SCENE`.
6. Prompt operator: "Is the job complete? (Y/N): "
7. If operator inputs 'Y' or 'y': return `True`.
8. Otherwise: return `False`.

### **complete\_call(state, call\_id)**

1. Search `state.call_queue` for call matching `call_id`.
2. If call not found or `call.call_status == CallStatus.COMPLETED`:
  - Print "[ERROR] Active pending Call ID not found."
  - Return `False`.
3. Set `call.call_status = CallStatus.COMPLETED`.
4. Set `call.unit_status = UnitStatus.CLEARED`.
5. For each officer in `call.assigned_officers.values()`:
  - Decrement `officer.active_cases` by 1.
  - Increment `officer.completed_cases` by 1.
6. Call `sort_queue(state)` to move completed call to end of queue list.
7. Return `True`.

### **sort\_queue(state)**

1. Re-order `state.call_queue` using key comparison function:
  - Primary Key: 0 if `status == CallStatus.PENDING` else 1.

- Secondary Key: -call.severity (descending priority for pending).
- Tertiary Key: call.call\_id (creation order).

2. Replace state.call\_queue with sorted list result.